CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II and IV Semester of BSc Physics Examination May 2018

PD 155 Chemistry – II

Date: 03/05/2018 Day: Thursday Time: 01.30 pm to 02.00 pm Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Que:1 Answer the following multiple choice questions. **[20]**

1 Phenol reacts with bromine water to give the precipitate of __________

(a) Bromophemol (b) Dibromophenol

(c) Tribromophenol (d) None of these

2 Which of the following reagents cannot be used to prepare an alkyl chloride from an alcohol?

(a) $HCl + ZnCl_2$ (b) $SOCl_2$

(c) NaCl (d) PCl_5

3 The reaction of ethyl bromide with sodium ethoxide give________

(a) Ethyl methyl ether (b) Diethyl ether

(c) Dimethyl ether (d) Above all

4 Haloform reaction is diagnostic test for ___________

(a) –CHO (b) -C=O (c) $-COCH_3$ (d) all of these

5 Iodobenzene heated with copper to form___________

(a) Biphenyl (b) Biiodophenyl

(c) Trphenyl (d) None of these

6 Ethers react with cold H_2SO_4 to form

(a) Alkenes (b) Zwitterions

(c) Alkoxides (d) Oxonium salts

7 Lucas test is used to determine the type of ________

(a)Acids (b) Alcohols (c) Ketones (d) Esters

8 Hydroboration-oxidation of propene gives _________

(a) Isopropyl alcohol (b) n-propyl alcohol

(c) Isobutyl alcohol (d) Tert-Butyl alcohol

9 Anisole is formed when phenol is treated with_________

(a) $CH_3I/NaOH$ (b) $CH_3CH_2I/NaOH$

(c) $CHCl_3/NaOH$ (d) Acetic anhydride

10 Acetone reacts with HCN to form cyanohydrins. It is an example of _____

(a) Electrophilic addition (b) Electrophilic substitution

(c) Nucleophilic addition (d) None of these

11 The study of the flow of heat or any other form of energy into or out of a system undergoing physical or chemical change is called_____

(a) Thermochemistry (b) Thermo kinetics

(c) Thermodynamics (d) Thermochemical studies

12 A system that can transfer neither matter nor energy to and from its surroundings is called

(a) A closed system (b) An isolated system

(c) An open system (d) A homogeneous system

13 Zinc granules reacting with dilute hydrochloric acid in an open beaker constitutes______

(a) An isolated system (b) An open system

(c) A closed system (d) A heterogeneous system

14 A system in which state variables have constant values throughout the system is called in a state of_____

(a) Equilibrium (b) Non-equilibrium

(c) Isothermal equilibrium (d) None of these

15 Which of the following relations is applicable to adiabatic expansion of an ideal gas?

(a) $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$ (b) $P_1V_1^{\gamma} = P_2V_2^{\gamma}$ (c) $P_1V_1 = P_2V_2$ (d) none of these

16 When a salt is added to a solution of another salt having a common ion, the degree of dissociation α ______

(a) Increases (b) Remain same

(c) Decreases (d) All of above

17 Ostwald's dilution law is applicable to______

(a) All electrolytes (b) Strong electrolytes

(c) Weak electrolytes (d) None of these

18 A weak acid together with a salt of the same acid with a strong base. These are called_____

(a) Acidic buffer (b) Basic buffer

(c) Both (d) None of these

19 The fraction of the amount of the electrolyte in the solution present as a free radical is called_______

(a) Dissociation (b) Precipitation

(c) Degree of dissociation (d) Degree of precipitation

20 The degree of dissociation of an electrolyte depends upon_____

(a) Nature of solute (b) Nature of solvent

(c) Concentration of solute (d) All of these

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PD 155 Chemistry – II

Date: 03/05/2018 **Day: Thursday** **Time: 02.00 pm to 04.30 pm** **Maximum Marks: 50**

Important Instructions:

- **Section I and II must be attempted in TWO ANSWER SHEET.**
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION -I [20]

Q - II

[A] Give the synthesis of crotonaldehyde and mesityl oxide. **[04]**

OR

[A] Write a note on: Reimer-Tiemann reaction. **[04]**

[B] Write a note on: Cannizzaro reaction. **[03]**

[C] Write a note on: Reformatsky reaction. **[03]**

Q - III

[A] Write a note on: Kolbe reaction. **[04]**

[B] Why the aryl halides are less reactive towards the nucleophilic displacement reaction? **[03]**

[C] Give the mechanism of sulphonation of benzene. **[03]**

OR

[C] Give the synthesis of aldehyde and ketone from the geminal dihalides. **[03]**

SECTION -II [30]

Q - IV

[A] Define heat capacity at constant pressure (C_P), heat capacity at constant volume (C_v) and derive the relation $C_P - C_V = R$. **[04]**

OR

[A] Derive the equation $P_1V_1^{\gamma} = P_2V_2^{\gamma}$ for adiabatic expansion of an ideal gas. **[04]**

[B] Explain the Kirchhoff's law in detail. **[03]**

[C] Explain common-ion effect by giving suitable examples. **[03]**

Q - V

[A] Write two preparation methods for phenol. **[04]**

[B] Write the oxidation reaction of 1°, 2°, 3° alcohol. **[03]**

[C] Explain the Williamson ether synthesis by giving suitable example. **[03]**

OR

[C] Write the Friedel-crafts alkylation reaction and explain its mechanism. **[03]**

Q - VI

[A] Explain the Debye-Huckel theory. **[04]**

[B] Derive the Hendersion-Hasselbalch equation for buffer solution. **[03]**

[C] Write the factors which influence the degree of dissociation. **[03]**

OR

[C] Calculate the value of ΔE and ΔH on heating 64.0 g of oxygen from 0 °C to 100ºC. C_v and C_p on an average are 5.0 and 7.0 cal mol^{-1} $degree^{-1}$. **[03]**